Mathematical Logic

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Outline

1. Introduction

2. First-order logic

Short History of Logic

- Logic is the study of reasoning.
- The first recorded use of logic was by Aristotle (384–322 BC).
- Aristotle explored the concept of arguing from premises to conclusions. He calls logic conclusions syllogisms.

Syllogism

A syllogism is a statement in which certain things [the premises] are asserted, and something else [the conclusion] necessarily follows from what is asserted. By the last sentence, I mean that the premises result in the conclusion, and by that, I mean that no additional premise is required to make the conclusion unavoidable.

Example: If all humans are mortal, and Socrates is a human, then Socrates is mortal.

Propositional Logic

Boole (1815–1864) introduced propositional logic in the 19th century.

Applications of Logic

- Electronic circuit design
- Logic programming (e.g. Prolog)
- Expert systems (form of AI)
- Databases (e.g. SQL using first-order logic)
- Formal verification of software

Language of first-order logic

A language $\mathscr{L}$ of first-order logic consists of the following components:

- Variable symbols: $x_1, x_2, \ldots$
- For each $n \in \mathbb{N}$, a set of n-ary function symbols: $f_0, f_1, \ldots$ The 0-ary function symbols are called constant symbols.
- For each $n \in \mathbb{N}$, a set of n-ary predicate symbols: $p_0, p_1, \ldots$ The 0-ary predicate symbols are the constants $\top$ (for **true**) and $\bot$ (for **false**).
- special symbols: $\neg$ (negation), $\wedge$ (conjunction), $\vee$ (disjunction), $\rightarrow$ (implication), $\leftrightarrow$ (equivalence), $\forall$ (universal quantification), $\exists$ (existential quantification), and parentheses.

Terms

The set of terms of $\mathscr{L}$ is defined inductively as follows:

- Each variable is a term.
- If $t_1, \ldots, t_n$ are terms and f is an n-ary function symbol, then if $f(t_1, \ldots, t_n)$ is a term.

Variables in terms

We define a function var : Terms $\rightarrow$ Variables that maps each term to the set of variables occurring in it. The function is defined as follows:

- $var(x) = \{x\}$ for each variable x.
- $var(f(t_1, \ldots, t_n)) = var(t_1) \cup \ldots \cup var(t_n)$.

Formulas

The set of formulas of $\mathscr{L}$ is defined inductively as follows:

- If $t_1, \ldots, t_n$ are terms and p is an n-ary predicate symbol, then if $p(t_1, \ldots, t_n)$ is a formula.
- If φ is a formula, then if $\neg\varphi$ is a formula.
- If φ_1 and φ_2 are formulas, then if $\varphi_1 \wedge \varphi_2$, $\varphi_1 \vee \varphi_2$, $\varphi_1 \to \varphi_2$, and $\varphi_1 \leftrightarrow \varphi_2$ are formulas.
- If φ is a formula and x is a variable, then if $\forall x.\varphi$ and $\exists x.\varphi$ are formulas.

An example of a formula is $\forall x.\exists y.p(x, y) \to \neg q(y)$.

Interpretations

An interpretation $\mathcal{M}$ of $\mathscr{L}$ consists of the following components:

- A non-empty set D called the domain of $\mathcal{M}$.
- For each n-ary function symbol f of $\mathscr{L}$, a function $f^{\mathcal{M}} : D^n \to D$.
- For each n-ary predicate symbol p of $\mathscr{L}$, a relation $p^{\mathcal{M}} \subseteq D^n$.

Interpretations of Terms

Let $\mathcal{M}$ be an interpretation for our first-order language. An assignment σ of values to variables, i.e., $\sigma : Variables \to D$.
The value of a term t under σ is denoted by $t^{\mathcal{M}}[\sigma]$ and defined as follows:

- If $t = x$ for a variable x, then $t^{\mathcal{M}}[\sigma] = \sigma(x)$.
- If $t = f(t_1, \ldots, t_n)$, then $t^{\mathcal{M}}[\sigma] = f^{\mathcal{M}}(t_1^{\mathcal{M}}[\sigma], \ldots, t_n^{\mathcal{M}}[\sigma])$.

Validity of Formulas under Interpretations

We say an assignment σ satisfies a formula φ under an interpretation $\mathcal{M}$, denoted by $\mathcal{M}, \sigma \models \varphi$, iff the following conditions hold:

- $\varphi = p(t_1, \ldots, t_n)$, then if $(t_1^{\mathcal{M}}[\sigma], \ldots, t_n^{\mathcal{M}}[\sigma]) \in p^{\mathcal{M}}$.
- $\varphi = \neg\psi$, then if $\mathcal{M}, \sigma \not\models \psi$.
- $\varphi = \psi_1 \vee \psi_2$, then if $\mathcal{M}, \sigma \models \psi_1$ or $\mathcal{M}, \sigma \models \psi_2$.
- $\varphi = \psi_1 \wedge \psi_2$, then if $\mathcal{M}, \sigma \models \psi_1$ and $\mathcal{M}, \sigma \models \psi_2$.
- $\varphi = \psi_1 \rightarrow \psi_2$, then if $\mathcal{M}, \sigma \models \psi_1$ implies $\mathcal{M}, \sigma \models \psi_2$.
- $\varphi = \psi_1 \leftrightarrow \psi_2$, then if $\mathcal{M}, \sigma \models \psi_1$ if and only if $\mathcal{M}, \sigma \models \psi_2$.
- $\varphi = \forall x.\psi$, then if $\mathcal{M}, \sigma[x \mapsto d] \models \psi$ for all $d \in D$.
- $\varphi = \exists x.\psi$, then if $\mathcal{M}, \sigma[x \mapsto d] \models \psi$ for some $d \in D$.

A formula φ is satisfiable if there exists an interpretation $\mathcal{M}$ and an assignment σ such that $\mathcal{M}, \sigma \models \varphi$.

Models

An interpretation $\mathcal{M}$ is a model of a formula φ, denoted by $\mathcal{M} \models \varphi$, if for all assignments σ, $\mathcal{M}, \sigma \models \varphi$.

A formula is satisfiable if it has a model, i.e., if there exists an interpretation $\mathcal{M}$ such that $\mathcal{M} \models \varphi$.

Validity

A formula φ is valid if for all interpretations $\mathcal{M}$ and all assignments σ, $\mathcal{M}, \sigma \models \varphi$.
We write $\models \varphi$ to denote that φ is valid.

Free Variables in Fomulas

The set of free variables of a formula φ, denoted by $FV(\varphi)$, is defined inductively as follows:

- $FV(p(t_1, \ldots, t_n)) = var(t_1) \cup \ldots \cup var(t_n)$.
- $FV(\neg\psi) = FV(\psi)$.
- $FV(\psi_1 \wedge \psi_2) = FV(\psi_1) \cup FV(\psi_2)$.
- $FV(\psi_1 \vee \psi_2) = FV(\psi_1) \cup FV(\psi_2)$.
- $FV(\psi_1 \rightarrow \psi_2) = FV(\psi_1) \cup FV(\psi_2)$.
- $FV(\forall x.\psi) = FV(\psi) \setminus \{x\}$.
- $FV(\exists x.\psi) = FV(\psi) \setminus \{x\}$.

Term Substitution

Let φ be a formula, x a variable, and t a term. The formula $\varphi[t/x]$ is obtained by replacing all occurrences of x in φ by t. The substitution is defined inductively as follows:

- $(p(t_1, \ldots, t_n))[t/x] = p(t_1[t/x], \ldots, t_n[t/x])$.
- $(\neg\psi)[t/x] = \neg\psi[t/x]$.
- $(\psi_1 \wedge \psi_2)[t/x] = \psi_1[t.x] \wedge \psi_2[t/x]$.
- $(\psi_1 \vee \psi_2)[t/x] = \psi_1[t/x] \vee \psi_2[t/x]$.
- $(\psi_1 \to \psi_2)[t/x] = \psi_1[t/x] \to \psi_2[t/x]$.
- $(\forall y.\psi)[t/x] = \forall y.\psi[t/x]$ if $x \in FV(t)$.
- $(\exists y.\psi)[t/x] = \exists y.\psi[t/x]$ if $x \in FV(t)$.
- $(\forall x.\psi)[t/x] = \forall x.\psi$.
- $(\exists x.\psi)[t/x] = \exists x.\psi$.

So, $\varphi[t/x]$ represents the formular obtained by substituting every **free** occurrence of the variable x in φ by the term t.

Calculus

A calculus is a mechanism to prove formulas by applying rules.
A rule of a calculus has the form $\frac{\varphi_1,\ldots,\varphi_n}{\psi}$, where $\varphi_1, \ldots, \varphi_n$ are premises and ψ is the conclusion. The rule states that if $\varphi_1, \ldots, \varphi_n$ are derivable, then ψ is derivable.
We denote that a formula can be proved by a calculus by $\vdash \varphi$.
We can also denote that a formula φ is derivable from a set of formulas (premises) Γ by $\Gamma \vdash \varphi$.

Calculus of Natural Deduction

- inspired by the way humans reason
- goal is to prove a conclusion from a set of premises
- In a natural deduction proof the formula occurring at the root of the tree is called the conclusion, while the formulas at the leaves of the tree are its assumptions.
- In a natural deduction proof the assumptions can be of two kinds: *canceled* and *uncanceled*.
- When one starts building ones proof tree all assumptions are uncanceled, but in certain inferences one is allowed to cancel certain assumptions.

Rules of Calculus of Natural Deduction

$$\frac{A \qquad B}{A \wedge B} \wedge I$$

$$\frac{A \wedge B}{A} \wedge E_1 \qquad \frac{A \wedge B}{B} \wedge E_2$$

$$\frac{A}{A \vee B} \vee I \qquad \frac{A \vee B \quad A \vdash C \quad B \vdash C}{C} \vee E$$

$$\frac{A \vdash B}{A \to B} \to I \qquad \frac{A \quad A \to B}{B} \to E$$

Rules of the Calculus of Natural Deduction

$$\frac{A \vdash B \quad A \vdash \neg B}{\neg A}\ \neg I \qquad \frac{A \quad \neg A}{\bot}\ \neg E$$

$$\frac{\bot}{A}\ \bot E \qquad \frac{A}{\neg\neg A}\ \neg\neg I\ (\neg\neg E \text{ analogously})$$

$$\frac{\Gamma \vdash A[u/x]\ (\text{wherefo } u \notin FV(\Gamma))}{\Gamma \vdash \forall x.A}\ \forall I \qquad \frac{\forall x.A}{A[t/x]\ t \text{ free for } x \text{ in } A}\ \forall E$$

Note: t free for x in A means: after inserting t for all free occurrences of x in A, no variable in t becomes bound by a quantifier in A.

Rules of the Calculus of Natural Deduction

$$\frac{A[t/x] \ (t \text{ free for } x \text{ in } A)}{\exists x.A} \exists I$$

$$\frac{\Gamma \vdash \exists x.A \qquad \Gamma, A[u/x] \vdash B \text{ where } u \notin FV(\Gamma) \cup FV(B) \text{ is a fresh variable}}{\Gamma \vdash B}$$

Example: I know that there is a Norwegian logician. I was able to infer from 'Skolem was Scandinavian and Logician'. Hence, from 'there exists a Norwegian logician', I can infer 'there exists a Scandinavian logician'.

Example Proof via the Calculus of Natural Deduction

We want to show: $((p \wedge q) \to q) \to r \vdash (p \wedge q) \to r$

$$\dfrac{\dfrac{\dfrac{\dfrac{[p \wedge q]^1}{q} \wedge E}{(p \wedge q) \to q} \to I \qquad ((p \wedge q) \to q) \to r}{r} \to E}{(p \wedge q) \to r} \to I_1$$

Sequent Calculus

- In sequent calculus, we have sequences $\Gamma \vdash \Delta$, where Γ and Δ are sets of formulas.
- The interpretation is that if all formulas in Γ are true, then at least one formula in Δ is true.
- Proofs in sequent calculus are performed backwards: We start from the conclusion and apply rules to derive the premises.

Sequent Calculus Rules

$$\frac{-}{\Gamma, \varphi \Rightarrow \varphi, \Delta}\ \text{Taut}$$

$$\frac{\Gamma \Rightarrow \Delta, \varphi \qquad \varphi, \Pi \Rightarrow \Lambda}{\Gamma, \Pi \Rightarrow \Delta, \Lambda}\ \text{Cut}$$

$$\frac{-}{\Gamma, \bot \Rightarrow \Delta}\ \bot \Rightarrow$$

$$\frac{-}{\Gamma \Rightarrow \Delta, \top}\ \Rightarrow \top$$

$$\frac{\Gamma \Rightarrow \Delta}{\varphi, \Gamma \Rightarrow \Delta}\ \text{Weakening left}$$

$$\frac{\Gamma \Rightarrow \Delta}{\Gamma \Rightarrow \Delta, \varphi}\ \text{Weakening right}$$

$$\frac{\varphi, \varphi, \Gamma \Rightarrow \Delta}{\varphi, \Gamma \Rightarrow \Delta}\ \text{Contraction left}$$

$$\frac{\Gamma \Rightarrow \Delta, \varphi, \varphi}{\Gamma \Rightarrow \Delta, \varphi}\ \text{Contraction right}$$

$$\frac{\Gamma, \varphi, \psi, \Pi \Rightarrow \Delta}{\Gamma, \psi, \varphi, \Pi \Rightarrow \Delta}\ \text{Exchange left}$$

$$\frac{\Gamma \Rightarrow \Delta, \varphi, \psi, \Lambda}{\Gamma \Rightarrow \Delta, \psi, \varphi, \Lambda}\ \text{Exchange right}$$

Sequent Calculus Rules

$$\frac{}{\Gamma, \bot \Rightarrow \Delta}\ \bot \Rightarrow \qquad \frac{}{\Gamma \Rightarrow \Delta, \top}\ \Rightarrow \top$$

$$\frac{\Gamma \Rightarrow \Delta, \varphi}{\Gamma, \neg\varphi \Rightarrow \Delta}\ \neg \Rightarrow \qquad \frac{\Gamma, \varphi \Rightarrow \Delta}{\Gamma \Rightarrow \Delta, \neg\varphi}\ \Rightarrow \neg$$

$$\frac{\Gamma, \varphi \Rightarrow \Delta \quad \Gamma, \psi \Rightarrow \Delta}{\Gamma, \varphi \vee \psi \Rightarrow \Delta}\ \vee \Rightarrow \qquad \frac{\Gamma \Rightarrow \Delta, \varphi, \psi}{\Gamma \Rightarrow \Delta, \varphi \vee \psi}\ \Rightarrow \vee$$

$$\frac{\Gamma, \varphi, \psi \Rightarrow \Delta}{\Gamma, \varphi \wedge \psi \Rightarrow \Delta}\ \wedge \Rightarrow \qquad \frac{\Gamma \Rightarrow \Delta, \varphi \quad \Gamma \Rightarrow \Delta, \psi}{\Gamma \Rightarrow \Delta, \varphi \wedge \psi}\ \Rightarrow \wedge$$

$$\frac{\Gamma \Rightarrow \Delta, \varphi \quad \psi, \Pi \Rightarrow \Lambda}{\varphi \to \psi, \Gamma, \Pi \Rightarrow \Delta, \Lambda}\ \to\Rightarrow \qquad \frac{\Gamma, \varphi \Rightarrow \Delta, \psi}{\Gamma \Rightarrow \Delta, \varphi \to \psi}\ \Rightarrow\to$$

Sequent Calculus Rules

$$\frac{\Gamma, \varphi[t/x] \Rightarrow \Delta}{\Gamma, \forall x.\varphi \Rightarrow \Delta} \forall \Rightarrow \qquad \frac{\Gamma \Rightarrow \Delta, \varphi[y/x]}{\Gamma \Rightarrow \Delta, \forall x.\varphi} \Rightarrow \forall$$

$$\frac{\Gamma, \varphi[y/x] \Rightarrow \Delta}{\Gamma, \exists x.\varphi \Rightarrow \Delta} \exists \Rightarrow \qquad \frac{\Gamma \Rightarrow \Delta, \exists x.\varphi, \varphi[t/x]}{\Gamma \Rightarrow \Delta, \exists x.\varphi} \Rightarrow \exists$$

In the quantifier rules, t is a term that is free for x in φ, and y is a 'fresh' variable, i.e., a variable that does not occur in Γ, Δ, or φ.
Alternatively, the rules can also be stated in the form

$$\frac{\Gamma \Rightarrow \Delta, \varphi}{\Gamma \Rightarrow \Delta, \forall x.\varphi} \Rightarrow \forall$$

Here, it must be guaranteed that x is not free in any formula in Γ or Δ.
The existential formula can be handled similarly.

Example Deduction: $\forall x.(A \to B) \Rightarrow A \to \forall x.B$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{A \Rightarrow A, B}\,\text{Taut} \quad \dfrac{}{A, B \Rightarrow B}\,\text{Taut}}{A, A \to B \Rightarrow B}\,{\to}\Rightarrow}{A, \forall x.(A \to B) \Rightarrow B}\,\forall \Rightarrow}{A, \forall x.(A \to B) \Rightarrow \forall x.B}\,\Rightarrow \forall}{\forall x.(A \to B) \Rightarrow A \to \forall x.B}\,\Rightarrow{\to}$$

Here, the application of $\Rightarrow \forall$ requires that x is not free in A.

Example of a Failing Deduction:

$\exists x.P(x) \land \exists x.Q(x) \Rightarrow \exists x.(P(x) \land Q(x))$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{P(x), Q(y) \Rightarrow P(x) \land Q(x)}{P(x), Q(y) \Rightarrow \exists x.(P(x) \land Q(x))} \Rightarrow \exists}{P(x), \exists x.Q(x) \Rightarrow \exists x.(P(x) \land Q(x))} \exists \Rightarrow}{\exists x.P(x), \exists x.Q(x) \Rightarrow \exists x.(P(x) \land Q(x))} \exists \Rightarrow}{\exists x.P(x) \land \exists x.Q(x) \Rightarrow \exists x.(P(x) \land Q(x))} \land \Rightarrow$$

Here, the deduction fails because the variable x is not fresh in the application of $\exists \Rightarrow$ and therefore the new variable y is introduced. However, then the deduction cannot be completed.

Soundness and Completeness of Sequent Calculus

- A calculus is sound if all provable formulas are valid, denoted by $\vdash \varphi \Rightarrow \models \varphi$.
- A calculus is complete if all valid formulas are provable, denoted by $\models \varphi \Rightarrow \vdash \varphi$.

The sequent calculus is sound and complete for first-order logic, i.e.,

- $\vdash \Gamma \Rightarrow \Delta$, then $\models \Gamma \Rightarrow \Delta$.
- $\models \Gamma \Rightarrow \Delta$, then $\models \Gamma \Rightarrow \Delta$.

Proof of soundness by induction on the structure of the formulas. Proof of completeness by constructing a Herbrand model.

Goedel's Incompleteness Theorem

- Goedel's first incompleteness theorem states that in any consistent formal system that is powerful enough to express arithmetic, there are true statements that cannot be proven.
- Goedel's second incompleteness theorem states that in any consistent formal system that is powerful enough to express arithmetic, the system cannot prove its own consistency.

Rice's Theorem

- Rice's theorem states that for any non-trivial property of partial functions, i.e., a property that is not true for all partial functions or not true for none, there is no algorithm that can decide whether a given program has that property.
- A property is non-trivial if there are two partial functions that are computable and one has the property and the other does not.

Halting Problem

The halting problem is the problem of determining, given a program and an input, whether the program will eventually halt when run with that input.
The halting problem is undecidable, i.e., there is no algorithm that can decide whether a given program halts on a given input.

Some Conclusions

- There are properties of programs that cannot be decided by an algorithm.
- There are properties of programs that cannot be verified by a formal system.
- It is impossible to generally prove behavioural equivalence of programs.